

Water Pump Project

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Introduction, Aims & Objectives

The aim of this project was to analyse the materials used in the flange, impeller and pump body of 6 different water pumps (A-F) made by different companies, and the processes used to manufacture each. Analysing the densities and material properties of each, I deduced the material used and identified the manufacturing process of each one through inspection of the surface finish under a microscope.

Each part operates under different mechanical, thermal and chemical conditions, meaning different parts will require different materials and processes. The parts play a vital role in the efficiency, durability and reliability of the pump, meaning the decision of which material and manufacturing process to use is an important one. Below are the tables containing the data collected for each of the 3 parts of the water pump, and the material identified for each.

Impeller:

Pump Impeller Ref	Mass (g) Immersed mass (g)	Density (kg/m ³)	Material considered	Material density published data	Impeller surface condition/colour manufacturing process	Outside diameter (mm)	Hole diameter (mm)	Comments
A	42.1 16.35	2575	Aluminium alloy	2626-2790	-silver -manufactured cast	54.83	11.94	-Lightweight -Non-magnetic
B	20.344 13.75	1479.6	uPVC + Glass composite	1300-1500	Injection moulded	55.45	17.74	-Non magnetic
C	41.11 15.55	2643.7	Aluminium alloy	2626-2790	-Silver -Powder metallurgy or die casting	55.35	15.88	-Delicate part -Likely die casted
D	138 16.7	8263	Brass	8430	-Gold -Powder Metallurgy	55.44	3.1	Non magnetic
E	26.9 13.75	2520	uPVC	1300-1500	-Blue -Smooth -Injection molding	55.37	13.65	-Non magnetic -Lightweight
F	43 16.2	2654.32	Aluminium alloy	2626-2790	-Grainy surface -Grey -Casted	56.11	11.74	-Non-magnetic -Lightweight

Flange:

Pump Ref	Mass (g) Immersed mass (g)	Density (kg/m ³)	Material considered	Material density published data	Impeller surface condition/colour manufacturing process)	Outside diameter (mm)	Hole diameter (mm)	Flange thickness Max/Min (mm)	M8 Thread Number of threads	Flange Profile
A	160.8 20.5	7844	Nickel Steel	7850	Turned on a lathe magnetic	58.07	15.73	8.77/11.94	7	Spigot
B	142.239 21.75	6539.7	Stainless Steel	7905	Powder metallurgy	57.92	15.50	7.58/11.43	7	Spigot
C	165.263 21.05	7850.83	Nickel Steel	7850	-Black coating -Milled	57.94	15.83	9.16/12.14	21	Spigot
D	233 31.2	7468	Cast Iron	7352	-Black magnetic -Turned on a lathe	58.04	6.82	13.12	9.5	Plain disc
E	259.2 22.2	7171.2	Cast Iron	7352	Silver/grey Magnetic Powder metallurgy	57.95	15.8	9.55/14.54	7	Spigot
F	160.2 20.65	7757.87	Nickel Iron	7822	Shiny silver Machined using lathe	15.84	15.84	8.80/11.89	7	Spigot

Pump body:

Pump Body Ref	Mass (g) Total	Cut sample Mass (g) & Immersed mass (g)	Density (kg/m ³)	Material considered	Material density published data	Body surface/ Colour manufacturing process
A	443.1	10.728 4.0	2682	Aluminium alloy	2626-2790	Manufacturing cast
B	498.7	10.525 3.895	2702.2	Aluminium alloy	2726-2790	Die casting
C	453.40	10.73 3.89	2758.35	Aluminium alloy	2626-2790	-Rough/grainy condition12 -Silver -Die casted/milled
D	507	13.8 5.2	2654	Aluminium alloy	2626-2790	-Silver -Die cast
E	501	10.535 3.889	2708.9	(Pure) Aluminium	2710	-Silver/grey -Grainy surface -Die casting
F	490	13.7 5.050	2712.87	(Pure) Aluminium	2710	-Less grainy than impeller -Grey -Die casted

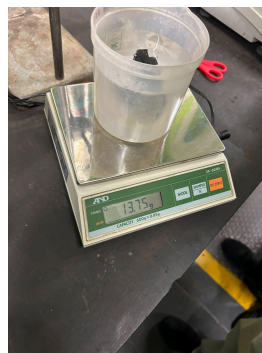
Materials & Process Summary Tables

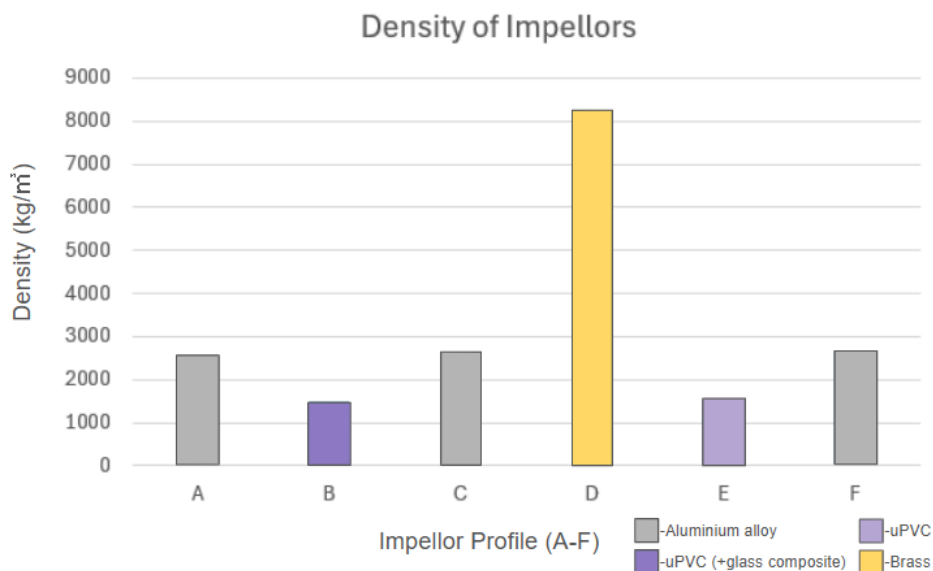
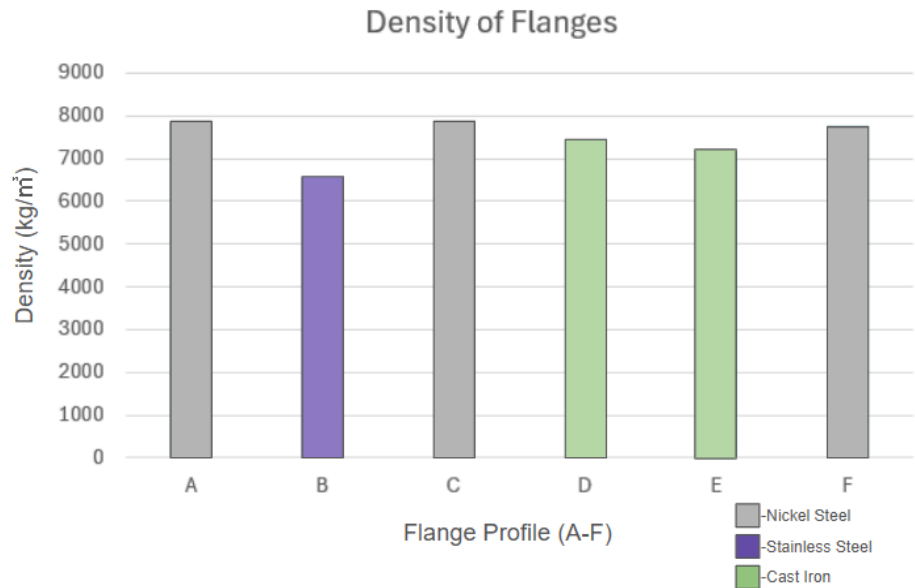
Flange Profile	Material	Manufacturing Process	Density (kg/m ³)
A	Nickel Steel	Turned on a lathe	7844
B	Stainless Steel	Powder metallurgy	6539.7
C	Nickel Steel	Milled	7850.83
D	Cast Iron	Turned on a lathe	7468
E	Cast Iron	Powder metallurgy	7271.2
F	Nickel Steel	Turned using a lathe	7757.87

Impeller	Material	Manufacturing Process	Density (kg/m ³)
A	Aluminium Alloy	Casting	2575
B	uPVC + Glass composite	Injection moulding	1479.6
C	Aluminium Alloy	Powder metallurgy or die casting	2643.7
D	Brass	Powder metallurgy	8263
E	uPVC	Injection moulding	1520
F	Aluminium Alloy	Casting	2654.32

Material Density Comparison Histograms

The data below is a visual representation of the density measurements we took for each part. We calculated these densities by measuring the mass of the part, and then the mass of the part submerged in water- this represented the volume of the part (in cm³), as water has a volume of 1g/cm³. This means the change in mass is therefore equal to the change in volume, and could be used to find the density of the part by dividing the original mass by the volume once it has been converted to m³ by multiplying it by 10⁻⁶.

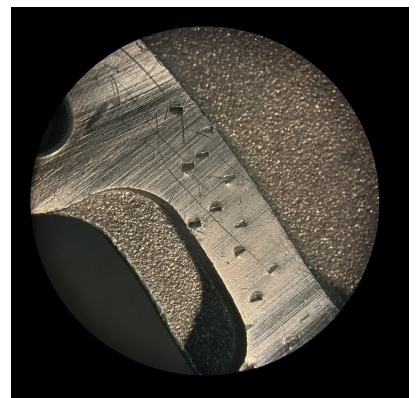




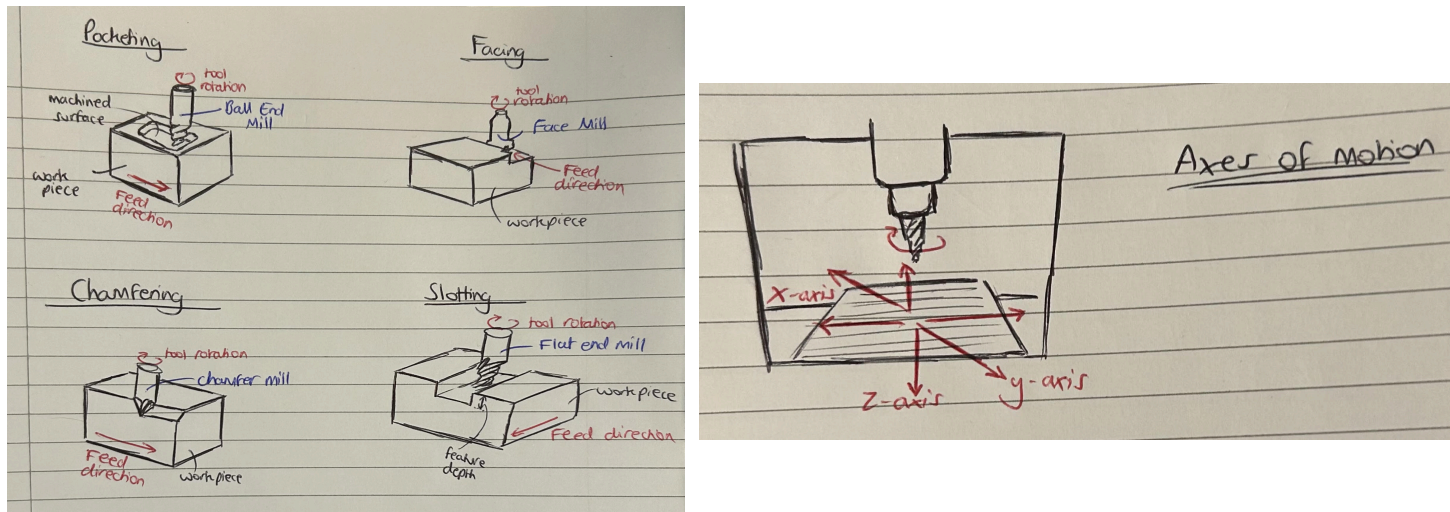
Primary Manufacturing Process

Pump body B was manufactured through milling. We identified this through a series of observations. First was the surface texture- the part had an extremely smooth and accurate finish, something which a form of milling is known for. Then we identified the lines on the surface of the body that were visible under the microscope. These came from the pitch of the mill bit used when machining the part, and are visible in the image on the right and on the page below.

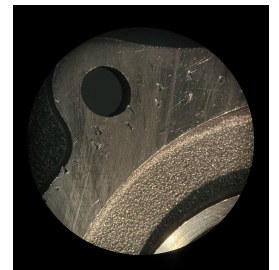
Milling involves the use of a rotating cutting tool to remove material from a workpiece, and is a common manufacturing method used to create precise shapes, holes, and surfaces in materials such as metals and plastics.



There are many different types of milling, each used for their own specific purposes. With this pump body, CNC Milling is most likely used due to the pump's intricate design and it being mass produced- CNC Milling has fast production times and highly accurate cuts, making it the ideal manufacturing process for this part. CNC Milling is a slightly more advanced version of regular milling- in regular milling, a rotating blade removes material from a part through the movement of the mill bit. This can take many forms, as illustrated below in Fig 3:



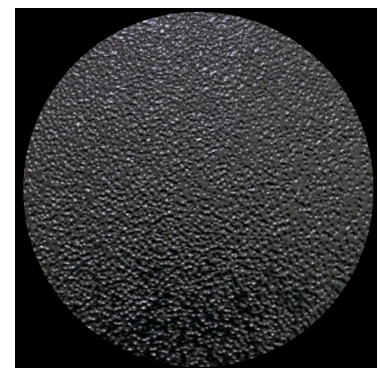
CNC Milling uses a combination of these milling techniques in 3 axes- x,y and z, with a series of precise cuts being made on each axis. CNC stands for Computer Numeric Control, the computer program needed to control the points at which the mill makes cuts in the metal- the software means these cuts are extremely accurate and consistent on each part made, and from a distance also give a smooth finish to the product. Up close under a microscope the lines from the mill bits are still visible, however for the accuracy and the rate of the process, it's an ideal manufacturing process used in mass production of parts such as the water pump body.



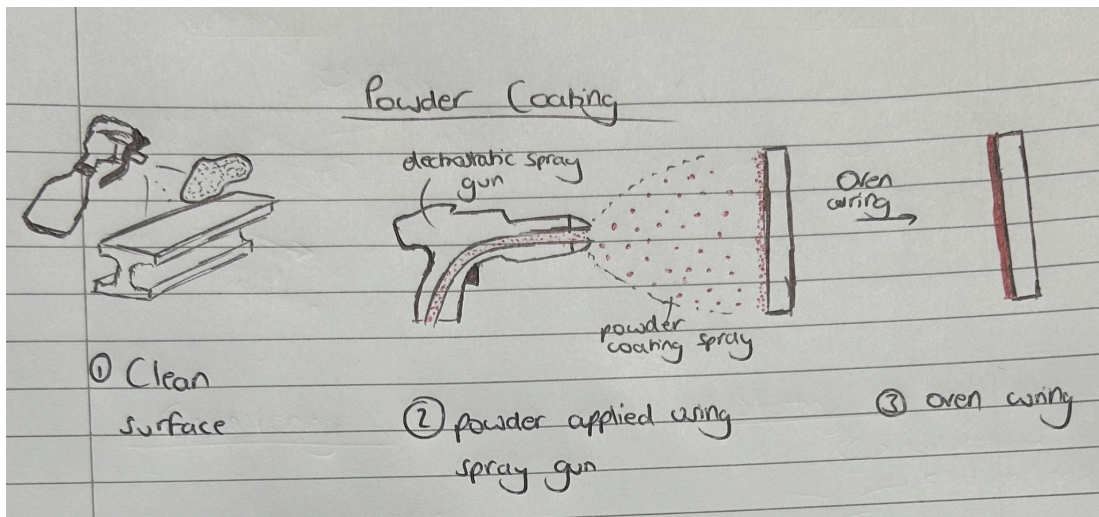
Secondary or Finishing Manufacturing Processes

We identified that Flange C was powder coated through similar methods. We first noticed how smooth and uniform the finish was, and how it had a slight rough texture to it. It was also very resistant to scratches, and under inspection in the microscope we were able to see the particles on the surface.

Powder coating is a dry finishing process where a fine powder is electrostatically sprayed onto a surface and then cured under heat to form a coating. The surface coated is usually metal, (just like the Nickel Steel that flange C is made of), and is cleaned and pre-treated to remove any rust or contaminants (1). The powder is then applied using a spray gun (2) which fires electrostatically charged powder particles onto the surface, where they then stick. The coated



part is then heated in an oven and cured (3), causing the powder to melt and fuse into a uniform layer. The part is then left to cool to solidify the finish.



(Fig 4)

Due to the process of heating to form a layer, the finish is extremely durable, resistant to any corrosion, chipping or fading. The powders don't contain any solvents so are also eco-friendly, and are fairly cheap as well making them a cost-efficient option for coating any parts. This makes it a widely used surface finish, used in a range of products from household appliances to car parts such as wheels and engine parts.